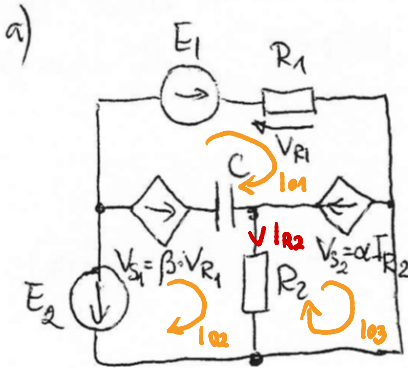


ĆWICZENIA 7

Zad. 1



$$V_{S1} = \beta \cdot V_{R1} = \beta \cdot R_1 \cdot I_{01}$$

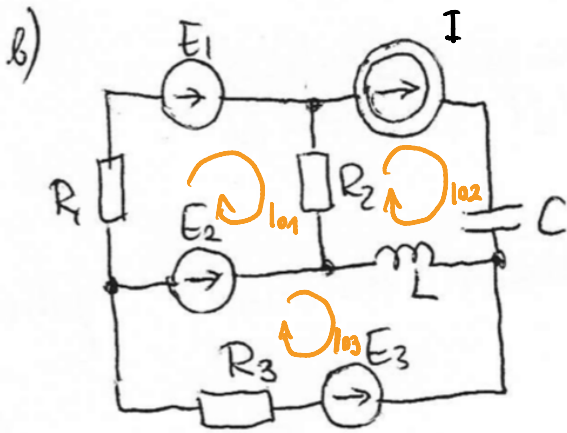
$$V_{S2} = \alpha \cdot I_{02} = \alpha (I_{02} - I_{03})$$

$$\begin{cases} (R_1 + Z_C) \cdot I_{01} - Z_C \cdot I_{02} - 0 \cdot I_{03} = E_1 + V_{S2} - V_{S1} \\ (R_2 + Z_C) \cdot I_{02} - Z_C \cdot I_{01} - R_2 \cdot I_{03} = V_{S1} - E_2 \\ R_2 \cdot I_{03} - R_2 \cdot I_{02} - 0 \cdot I_{01} = -V_{S2} \end{cases}$$

$$\begin{cases} (R_1 + Z_C) \cdot I_{01} - Z_C \cdot I_{02} - 0 \cdot I_{03} = E_1 + \alpha I_{02} - \alpha I_{03} - \beta \cdot R_1 \cdot I_{01} \\ (R_2 + Z_C) \cdot I_{02} - Z_C \cdot I_{01} - R_2 \cdot I_{03} = \beta \cdot R_1 \cdot I_{01} - E_2 \\ R_2 \cdot I_{03} - R_2 \cdot I_{02} - 0 \cdot I_{01} = -\alpha I_{02} + \alpha I_{03} \end{cases}$$

$$\begin{bmatrix} R_1 + \beta R_1 + Z_C & -\alpha - Z_C & \alpha \\ -Z_C - \beta R_1 & R_2 + Z_C & -R_2 \\ 0 & \alpha - R_2 & R_2 - \alpha \end{bmatrix} \cdot \begin{bmatrix} I_{01} \\ I_{02} \\ I_{03} \end{bmatrix} = \begin{bmatrix} E_1 \\ -E_2 \\ 0 \end{bmatrix}$$

Zad. 2

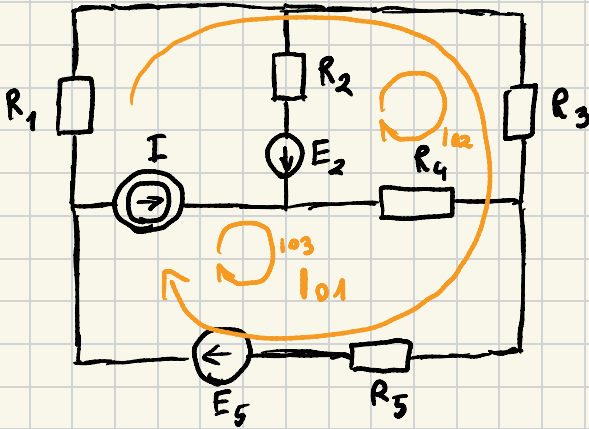


$$I_{02} = I$$

$$\begin{cases} (R_1 + R_2) \cdot I_{01} - R_2 \cdot I_{02} = E_1 - E_2 \\ (R_3 + Z_L) \cdot I_{03} - Z_L \cdot I_{02} = E_2 - E_3 \end{cases}$$

$$\begin{cases} (R_1 + R_2) \cdot I_{01} = E_1 - E_2 + I R_2 \\ (R_3 + Z_L) \cdot I_{03} = E_2 - E_3 + I Z_L \end{cases}$$

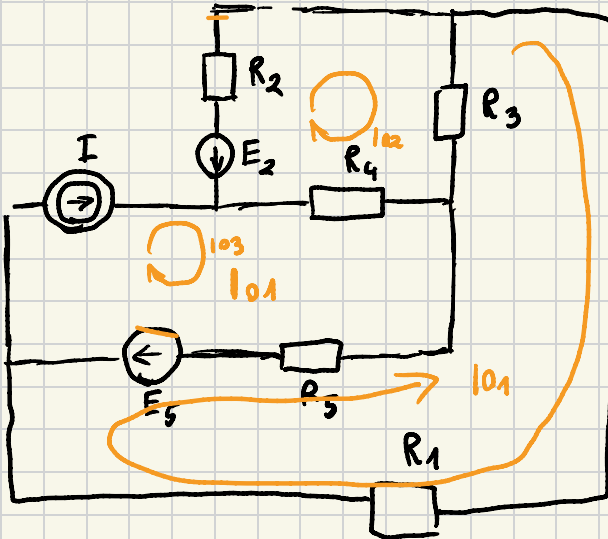
Zad. 3



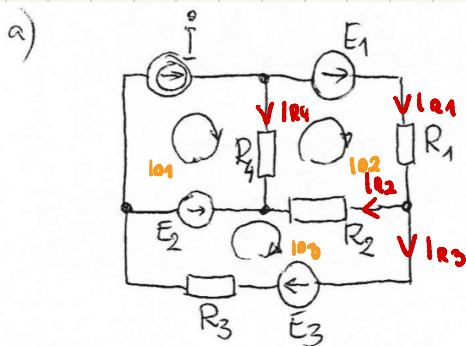
$$I_{03} = I$$

HUH?

$$\begin{cases} I_{02}(R_2 + R_4 + R_3) - R_3 \cdot I_{01} - R_4 \cdot I_{03} = -E_2 \\ I_{01}(R_1 + R_3 + R_5) - R_3 \cdot I_{02} - R_5 \cdot I_{03} = E_5 \end{cases}$$



Zad. 4



$$i(t) = 4\sqrt{2} \sin(\omega t)$$

$$e_1(t) = 20\sqrt{2} \sin(\omega t + 80^\circ)$$

$$e_2(t) = 10\sqrt{2} \sin \omega t$$

$$e_3(t) = 20 \sin(\omega t + 45^\circ)$$

$$R_1 = R_2 = R_3 = R_4 = 10 \Omega$$

$$I = 4 [A]$$

$$E_1 = j20$$

$$E_2 = 10$$

$$E_3 = 10\sqrt{2} \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j \right) = 10 + 10j$$

$$I_{01} = I = 4 A$$

$$\begin{cases} (R_1 + R_2 + R_4) \cdot i_{02} - R_4 \cdot i_{01} - R_2 \cdot i_{03} = E_1 \\ (R_2 + R_3) \cdot i_{03} - R_2 \cdot i_{02} - 0 \cdot i_{01} = E_3 + E_2 \end{cases}$$

$$\begin{cases} (10 + 10 + 10) \cdot i_{02} - 10 \cdot 4 - 10 \cdot i_{03} = j20 \\ (10 + 10) \cdot i_{03} - 10 \cdot i_{02} = 10 + 10j + 10 \end{cases}$$

$$\begin{cases} 30i_{02} - 10i_{03} = 40 + j20 \quad || : 10 \\ 20i_{03} - 10i_{02} = 20 + j10 \quad || : 10 \end{cases}$$

$$\begin{cases} 3i_{02} - i_{03} = 4 + j2 \quad || \cdot 2 \\ 2i_{03} - i_{02} = 2 + j \end{cases}$$

$$\begin{cases} 6I_{02} - 2I_{03} = 8 + j4 \end{cases}$$

$$\begin{cases} 2I_{03} - I_{02} = 2 + j \end{cases}$$

$$6I_{02} - I_{02} = 8 + j4 + 2 + j$$

$$5I_{02} = 10 + j5 \quad || : 5$$

$$I_{02} = 2 + j$$

$$2I_{03} - 2 - j = 2 + j$$

$$2I_{03} = 4 + 2j$$

$$I_{03} = 2 + j$$

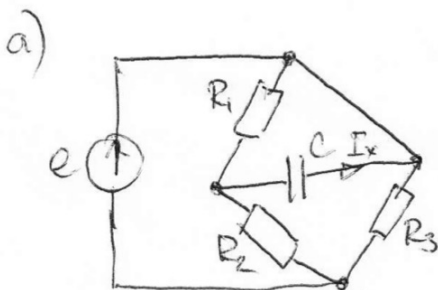
$$I_{R2} = I_{02} - I_{03} = 2 + j - (2 + j) = 0$$

$$I_{R1} = I_{02} = 2 + j$$

$$I_{R3} = I_{03} = 2 + j$$

$$I_{R4} = I_{01} - I_{02} = 4 - (2 + j) = 2 - j$$

Zad. 5

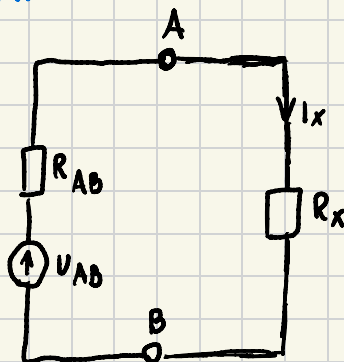


$$e(t) = 20\sqrt{2} \sin(\omega t - 90^\circ)$$

$$R_1 = 10\Omega, R_2 = 20\Omega, R_3 = 30\Omega$$

$$X_c = \frac{1}{\omega C} = 5\Omega$$

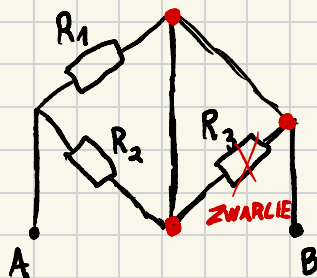
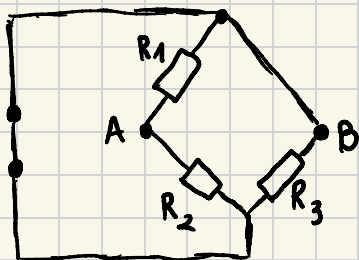
TW. THEVENINA



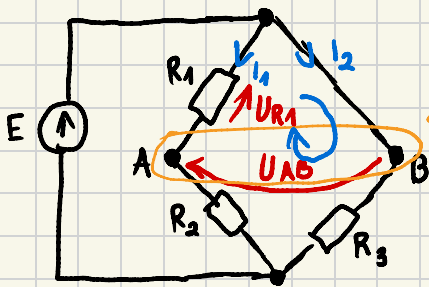
$$E = -j20 \quad R_1 = 10\Omega \quad R_2 = 20\Omega \quad R_3 = 30\Omega$$

$$X_c = \frac{1}{\omega C} = 5\Omega$$

$$R_X = Z_c = -jX_c = -j5$$



$$R_{AB} = \frac{R_1 \cdot R_2}{R_1 + R_2} = \frac{10 \cdot 20}{10 + 20} = \frac{200}{30} = 6,67 \Omega$$



Traktujemy, jakby tej
gałęzi nie było licząc I_1 i I_2

$$I_1 = \frac{E}{R_1 + R_2}$$

$$I_2 = \frac{E}{R_3}$$

$$U_{AB} + U_{R1} = 0$$

$$U_{AB} = -U_{R1}$$

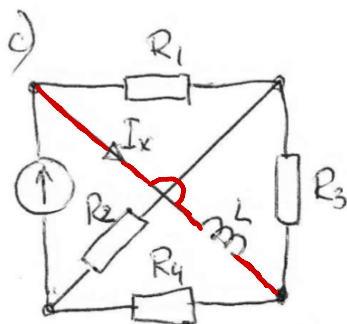
$$U_{AB} = 6,67j$$

$$I_1 = \frac{-j20}{30} = -0,67j$$

$$U_{R1} = I_1 \cdot R_1 = -0,67j \cdot 10 = -6,67j$$

$$I_x = \frac{U_{AB}}{R_{AB} + R_x} = \frac{6,67j}{6,67 - j5} = \dots$$

Zad. 6

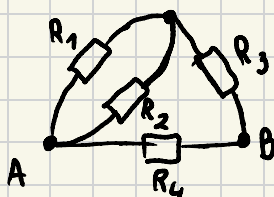
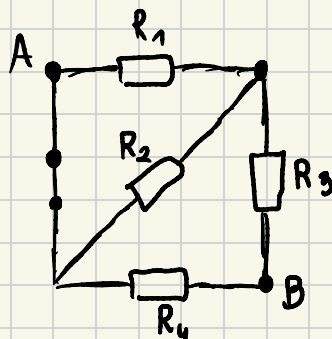


$$e(t) = 20 \sin(\omega t + 45^\circ)$$

$$R_1 = 10 \Omega, R_2 = 10 \Omega$$

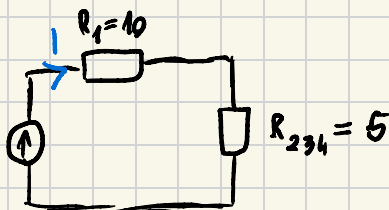
$$R_3 = R_4 = 5 \Omega, \omega L = 10 \Omega$$

$$E = \frac{20}{\sqrt{2}} \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} j \right) = 10 + 10j$$



$$R_{12} = \frac{R_1 \cdot R_2}{R_1 + R_2} = \frac{10 \cdot 10}{10 + 10} = \frac{100}{20} = 5 \Omega$$

$$R_{AB} = \frac{10 \cdot 5}{15} = \frac{50}{15} = 3,33 \Omega$$

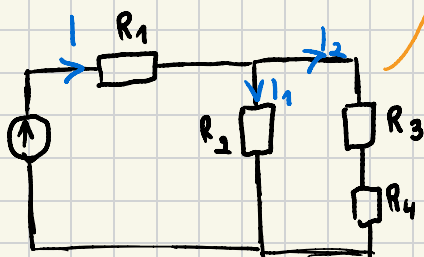
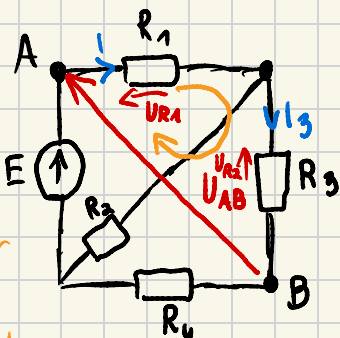


$$I = \frac{E}{R_1 + R_{2,3,4}} = \frac{10 + 10j}{15} = 0,67 + 0,67j$$

$$U_{R_{2,3,4}} = 5 \cdot (0,67 + 0,67j) = 3,33 + 3,33j$$

$$U_{R1} = 10 \cdot (0,67 + 0,67j) = 6,67 + 6,67j$$

$$I_3 = \frac{U_{R_{2,3,4}}}{R_3} = \frac{3,33 + 3,33j}{5} = 0,67 + 0,67j$$



$$U_{R3} = 5(0,67 + 0,67j) = 3,33 + j3,33$$

$$U_{AB} - U_{R1} - U_{R3} = 0$$

$$U_{AB} = U_{R1} + U_{R3} = 10 + 10j \text{ [V]}$$

$$I_X = \frac{U_{AB}}{R_{AB} + R_X} = \frac{10 + 10j}{3,33 + j10}$$